Team 2

P3 Process Book: Pre-Trained Models

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AGENDA

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Initial Capability Brainstorm - Amy

CAPABILITY: WRITE

Description based on image and keywords - **retail services** - automatically generate product descriptions for clothing brands (fast fashion particularly), furniture (ikea, costco, etc), or small businesses (etsy) based off photos, and existing data keywords such as product specs or manufacturing details

Character's response based on dialogue - **ed tech** - allow students to interact more meaningfully with historical literature by speaking to characters (in character) based on training from the historical text. For example, speaking to macbeth in character to gain more insight into how he thinks or why he takes his actions.

Character's response based on dialogue - **customer service** - train agents as "characters" with the goal to assist consumers, have them respond accordingly to assist online shoppers

Write a description based on keywords - **blogging/personal use** - help people write blog articles/substacks faster, train off their previous data or writing style in order to generate scheduled blog posts and post them with less user effort or involvement.

Initial Capability Brainstorm - Amy

CAPABILITY: WRITE

Character's response based on dialogue - **script/scenewriting** - In books/plays or movies, particularly adaptations (turning a book into a play or movie script), upload the book and the character descriptions, what can be cut or what must be kept, and generate scripts for plays or movies from the book quickly and without losing sight of the original plot or message.

CAPABILITY: TRANSLATE

Translate into a description from computational code - **Software/Corporate Documentation** - Take in tooling or code created by workers and generate descriptions and formatting for the code in order to more seamlessly integrate it into larger codebases.

Translate into computational code from a description - **Software/Corporate Documentation** - Take in code from previous models or outside/open source/acquired products and read and understand it, then generate documentation on how to run or use it and how to integrate it into an existing codebase

Initial Capability Brainstorm - Amy

CAPABILITY: TRANSLATE

Translate from computational code - **Updating codebases** - take in old or deprecated code that no longer runs due to things like unsupported version issues or library changes and cross-reference published updates and documentation to update the code to run currently (e.g. turn a python 2.7 program that doesn't run into a python 3.16 that does)

CAPABILITY: WRITE

Description based on image and keywords - **accessibility** - automatically generate text descriptions for images on the internet that can be read to people with visual impairments or disabilities, as well as potentially generating assistive movie voiceovers with much less effort

Description based on image and keywords - **social media** - suggest captions for instagram posts or youtube videos with the most optimal word count, trending hashtags, algorithmically optimized to do as well as possible on the platform

Concept Cards - Amy

Concept Card					
Title	Interactive literature for education				
Description	Character's response based on dialogue - ed tech - allow students to interact more meaningfully with historical literature by speaking to characters (in character) based on training from the historical text. For example, speaking to macbeth in character to gain more insight into how he thinks or why he takes his actions.				
Who are the...		*What should the system..*		*What do we know?*	
Payer (Customer)	Public schools/school systems/homeschooling programs	**Do / Act**	Greet the user in-character as a character from a book/play/movie, respond to the user's questions in character and discuss internal motivations, thoughts, etc.	**Data 1**	There exists pretrained models that can be set to specific character settings to react and interact with the user according to the uploaded context in-character (c.ai, customer service bots, etc)
End User	Kids in classrooms			**Data 2**	There are many analysis papers on most famous (taught in-school) literature that can further be added to the model in order to give in-depth responses
Company or Department	EdTech companies (e.g., curriculum platforms, LMS providers) School districts / curriculum design departments Educational publishers (Pearson, McGraw Hill)	**Infer / Reason**	Students struggle with engagement and comprehension of classical literature due to archaic language and lack of relatability Conversational interaction increases retention and emotional connection to material Teachers need scalable tools to personalize learning without increasing workload Existing literature analysis tools are passive; this introduces active learning	**Data 3**	Large language models are strong at role-playing and maintaining stylistic consistency when prompted or fine-tuned on specific texts
Impacted Individuals	Students (primary) Teachers (lesson planning + facilitation) Parents (homeschool environments) Curriculum designers			**Data 4**	Educational research shows active learning (discussion, simulation) significantly improves comprehension and retention compared to passive reading

Process | Conceptual | Craft & Professional Practice | Individual Reflection

Concept Cards - Amy

Does Anyone Want This?		
User Value: How much value do we provide to the end-user? (e.g., getting the job done, status enhancement, convenience, usability, newness, customization, performance...)	It can create an engaging method for students to interact with otherwise difficult to understand literature, although it should be treated as a supplement and not a replacement - Medium	Medium
Customer Value: How much value do we deliver to the payer? (e.g., getting the job done, status enhancement, convenience, usability, newness, customization, performance...)	Our product is not a necessity, students can get by without it, but it could serve as a more fun and engaging way for students to interact with literature	Medium
Service Value (Revenue): To what degree will this application generate revenue?	If we make it a pay-to-use subscription for big customers like school districts, we stand to generate a very large profit	High
Service Value (Non-revenue): How much non-financial value does this concept add to our service? (e.g. employee engagement, corporate social responsibility, raising awareness, etc)	It helps with literacy and education,	Medium
Adoption + Frequency of Use: How many new and current users will use this service, and how frequently?	Students, at least a few times a year during specific units like shakespeare perhaps	Medium
Risk to User/Stakeholder: How likely is this to cause harm to end-users or stakeholders? (e.g., their physical or psychological safety, civil liberties or rights, or economic opportunity)	Risks misinterpretation of text, potential learning of bad values	Low

Can This Be Built?		
Data Quality: Is there enough high-quality data to achieve 'good enough' model performance? (e.g., think about accuracy, biases, completeness ... of the data)	Public domain texts + analysis + summaries widely available	High
Minimum Performance: How well does the model need to perform for users to experience value?	Must maintain consistent character voice and avoid hallucinating incorrect plot details	Excellent
Data Bias: Is the data Biased? Will the system perform differently for certain groups or scenarios, potentially causing significant harm?	Bias toward canonical Western literature; may underrepresent diverse texts	Medium
Error Frequency: How frequently would errors occur?	Occasional misinterpretations or out-of-character responses	Medium
Error Severity: How severe would an error be?	Risks misinterpretation of text, potential learning of bad values	Low
Bad Actors: How easily could bad actors misuse the system to cause harm? (e.g., leading to harming data quality, security breaches, or monetary loss)	All responses surveyed by teachers and educators	Low
In Summary: Explain the rationale behind your ratings in this category. Be concise. Add a few words for a red flag. (e.g. "people will misuse", "teachers will feel uncomfortable")	Strong educational upside but not essential Risk of over-reliance replacing critical thinking High build cost justified only if integrated into broader curriculum systems Red flag: "students may rely on AI instead of reading"	

Concept Cards - Amy

Concept Card					
Title					
Description	Translate into a description from computational code - Software/Corporate Documentation - Take in tooling or code created by workers and generate descriptions and formatting for the code in order to more seamlessly integrate it into larger codebases.				
Who are the...		*What should the system..*		*What do we know?*	
Payer (Customer)	Tech companies (startups and large enterprises) Engineering teams / CTO orgs	**Do / Act**	Parse codebases (functions, classes, dependencies) Generate: Inline comments README files API documentation Update documentation dynamically as code changes Suggest missing documentation or inconsistencies	**Data 1**	LLMs already perform strong code summarization (GitHub Copilot, GPT-based tools)
End User	Software engineers New hires / onboarding engineers Technical writers			**Data 2**	Large open-source repositories provide abundant training data for code + documentation pairs
Company or Department	Engineering teams DevOps / Platform teams Developer productivity teams	**Infer / Reason**	Documentation is consistently neglected due to time constraints Poor documentation slows onboarding and increases bugs Engineers prefer automated tooling over manual writing Companies lose productivity due to knowledge silos	**Data 3**	Static analysis tools can extract structure (ASTs) to improve accuracy of generated docs
Impacted Individuals	Developers (time savings) Technical writers (role shift toward editing) Product managers (clearer system understanding)			**Data 4**	Companies already pay for developer productivity tools (high willingness to pay)

Concept Cards - Amy

Does Anyone Want This?		
User Value: How much value do we provide to the end-user? (e.g., getting the job done, status enhancement, convenience, usability, newness, customization, performance...)	Documentation for programming is very necessary - this can replace a large amount of otherwise tedious manual work - High	High
Customer Value: How much value do we deliver to the payer? (e.g., getting the job done, status enhancement, convenience, usability, newness, customization,	Reduces engineering time spent on documentation Improves onboarding speed and team velocity Decreases technical debt	High
Service Value (Revenue): To what degree will this application generate revenue?	Large corporate offices will likely pay premium dollar for a feature like this	High
Service Value (Non-revenue): How much non-financial value does this concept add to our service? (e.g. employee engagement,	It improves efficiency as well as creates a necessary resource (documentation) and also can streamline the codebase changes and updates by being able to autogenerate new and integrated documentation	High
Adoption + Frequency of Use: How many new and current users will use this service, and how frequently?	Applicable for almost every worker or dev at a tech company or startup	High
Risk to User/Stakeholder: How likely is this to cause harm to end-users or stakeholders? (e.g., their physical or psychological safety, civil liberties or rights, or economic opportunity)	Incorrect documentation can mislead people or cause larger slowdowns that without the tool	Medium

Can This Be Built?		
Data Quality: Is there enough high-quality data to achieve 'good enough' model performance? (e.g., think about accuracy, biases, completeness ... of the data)	Massive availability of code + doc pairs (GitHub, internal repos)	High
Minimum Performance: How well does the model need to perform for users to experience value?	Must be highly accurate; incorrect documentation causes bugs and misuse	Excellent
Data Bias: Is the data Biased? Will the system perform differently for certain groups or scenarios, potentially causing significant harm?	Bias toward popular languages/frameworks	Low
Error Frequency: How frequently would errors occur?	Edge cases in complex or poorly written code	Medium
Error Severity: How severe would an error be?	incorrect documentation causes bugs and misuse	High
Bad Actors: How easily could bad actors misuse the system to cause harm? (e.g., leading to harming data	High-leverage task, very easy to misuse since nobody would really be checking if they were using this tool	High
In Summary: Explain the rationale behind your ratings in this category. Be concise. Add a few words for a red flag. (e.g. "people will misuse", "teachers will feel uncomfortable")	Strong ROI and clear enterprise demand Fits existing workflows (IDE integrations, CI/CD) High cost but justified by productivity gains Red flag: “incorrect docs can propagate errors at scale”	

Initial Capability Brainstorm - Davis

CAPABILITY: Rank

- Car Dealerships: Buyers can input their personality and what they're looking for so the AI can rank those accordingly with the salespeople and the dealership can assign them the most optimal readily-available salesperson
- For painting houses or buildings, users can put in the vibe of what they want and some color schemes they were thinking about, and AI can help rank them in order of what is most likely to achieve the desired output
- For grocery stores, they can use AI to rank produce by its ripeness, so that way only ripe (or soon to be ripe) items are on the shelf
- People can feed what they eat every day along with their mood, time of day, how much you liked the dish, and weather to an AI, which will then also take a menu of a restaurant and let you know which items you are most likely to enjoy
- For tutors, they can put practice tests and student answers and work into AI, and it can rank the foundational things that need to be focused on, or if everything is all good then what to do next

Initial Capability Brainstorm - Davis

CAPABILITY: Rank

- A company with support tickets/help requests will take them through a form, where AI can rank them based on criticality from a description, as well as potentially directing the most capable employee to the ticket

CAPABILITY: Identify

- Bike/Skateboard shops can ask customers for pictures of their bikes/skateboards before coming in so they make sure to order the parts ahead of time and be ready for repair when the customer comes in
- Interior decorators can put pictures into AI which will identify common styles in the picture and give suggestions on where to look for furniture and paints
- User uploads picture of an animal or bug to AI, which identifies what the animal or bug is and the best (tamest) and safest way to remove them
- AI can quickly scan reviews of restaurants to see what dishes people enjoyed or didn't enjoy so the chef knows what to improve
- Students and researchers can use AI to identify the argument made by a piece of literature or previous research. This way new researchers don't spend so much time doing background research and can get to make contributions

Concept Cards - Davis

Concept Card							
Title	Ranking personas based off of a description						
Description	Car Dealerships: Buyers can input their personality and what they're looking for so the AI can rank those accordingly with the salespeople and the dealership can assign them the most optimal readily-available salesperson						
Who are the...		*What should the system..*		*What do we know?*			
Payer (Customer)	Car Dealerships	**Do / Act**	The system will take in a description of the person lookir	**Data 1**	Previous sales data		
End User	Car Dealership salespeople			**Data 2**	Customer description		
Company or Department	Sales team	**Infer / Reason**	Given a description, the AI will look at previous sales wit	**Data 3**	salespeople description		
Impacted Individuals	Car salespeople & customers			**Data 4**	Salespeople availability		

Concept Cards - Davis

Does Anyone Want This?		
User Value: How much value do we provide to the end-user? (e.g., getting the job done, status enhancement, convenience, usability, newness, customization, performance...)	They get to sell more cars + increased revenue (HIGH)	High / Medium / Low
Customer Value: How much value do we deliver to the payer? (e.g., getting the job done, status enhancement, convenience, usability, newness, customization, performance...)	Refer above	High / Medium / Low
Service Value (Revenue): To what degree will this application generate revenue?	This will definitely generate new revenue, as sales can be very personality-based and this would fix that gap	High / Medium / Low
Service Value (Non-revenue): How much non-financial value does this concept add to our service? (e.g. employee engagement, corporate social responsibility, raising awareness, etc)	This would increase employee interaction and presence within the company, because they would be motivated to make more money through commision. It would also tell car customers to come to dealership A for a good experience buying a car	High / Medium / Low
Adoption + Frequency of Use: How many new and current users will use this service, and how frequently?	All dealerships would implement this immediately given the chance	High / Medium / Low
Risk to User/Stakeholder: How likely is this to cause harm to end-users or stakeholders? (e.g., their physical or psychological safety, civil liberties or rights, or economic opportunity)	If the descriptions of a given customer is not fleshed out well or provide an accurate representation, it could harm economic opportunity	High / Medium / Low
In Summary: Explain the rationale behind your ratings in this category. Be concise. Add a few words for a red flag. (e.g. "people will misuse", "teachers will feel uncomfortable")	Low risk, it just improves upon an existing system	

Can This Be Built?		
Data Quality: Is there enough high-quality data to achieve 'good enough' model performance? (e.g., think about accuracy, biases, completeness ... of the data)	The amount of high quality data would increase as it ran, but nonexistant so far (LOW)	High / Medium / Low
Minimum Performance: How well does the model need to perform for users to experience value?	Increase car sales marginally	Excellent / Good / Moderate
Data Bias: Is the data Biased? Will the system perform differently for certain groups or scenarios, potentially causing significant harm?	Could avoid phrenology by not accepting what the customer looks like	High / Medium / Low
Error Frequency: How frequently would errors occur?	Error rate would decrease as more data was input	High / Medium / Low
Error Severity: How severe would an error be?	Loss of car sale	High / Medium / Low
Bad Actors: How easily could bad actors misuse the system to cause harm? (e.g., leading to harming data quality, security breaches, or monetary loss)	Car customers could input their given personality, but not be looking to buy a car in the first place	High / Medium / Low
In Summary: Explain the rationale behind your ratings in this category. Be concise. Add a few words for a red flag. (e.g. "people will misuse", "teachers will feel uncomfortable")	Could be built and would get better with time, but window shoppers or customer moods v	

Should This Be Built?		
Build Costs: How expensive/time-consuming to build a dataset, develop a model, or fine-tune a pre-built model? (e.g., systems license, programming, data collection, employees, training, etc)	Unknown	High / Medium / Low
Run Costs: How expensive would it be to run? (e.g., electricity, frequency of updating and retraining the model, data storage, computing power requirements, etc)	Unknown	High / Medium / Low
Societal Harm: How likely is it that this will indirectly cause societal harm? (e.g., job loss, sustainability, etc)	Would be very hard for someone to be punished from this	High / Medium / Low
In Summary: Explain the rationale behind your ratings in this category. Be concise. Add a few words for a red flag. (e.g. "people will misuse", "teachers will feel uncomfortable")	I don't know/we weren't taught any way to make these judgements	

Judgment		
This idea is a:	Maybe	Keeper / Maybe / Show-Stopper

Concept Cards - Davis

Concept Card					
Title	Identifying damages				
Description	The system analyzes user-submitted images and recognizes specific damage or components so that repairs can be prepared before the item physically arrives.				
Who are the...		*What should the system..*		*What do we know?*	
Payer (Customer)	Car repair centers	**Do / Act**	Identify damage, suggest possible replacements, and fla	**Data 1**	Images of damaged car
End User	The engineers who will be maintaining and fixing things			**Data 2**	Vehicle details
Company or Department	Mechanics	**Infer / Reason**	N/A	**Data 3**	parts database/availability
Impacted Individuals	Car owners, mechanics, insurance companies,			**Data 4**	Past repair records

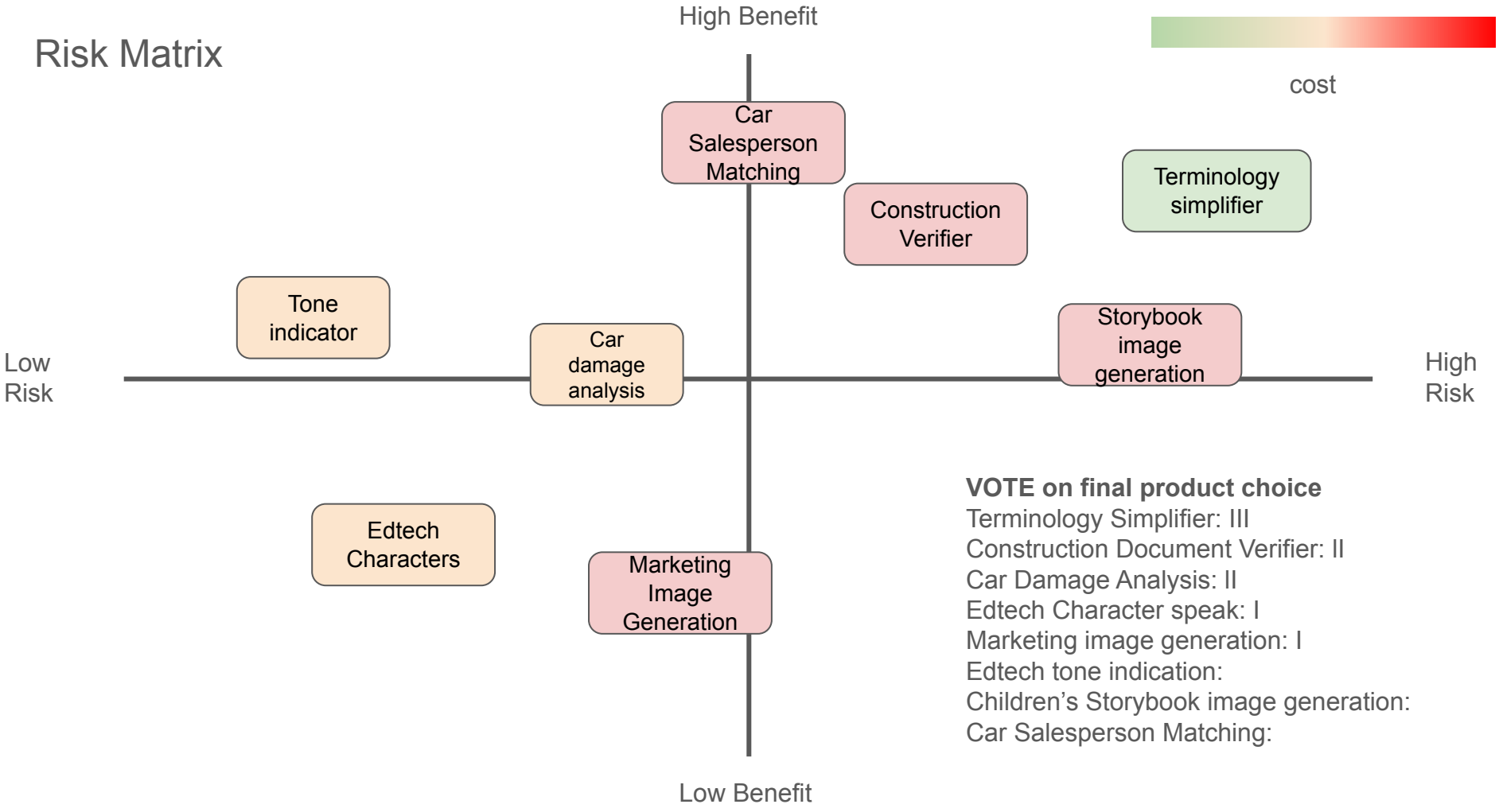
Concept Cards - Davis

Does Anyone Want This?		
User Value: How much value do we provide to the end-user? (e.g., getting the job done, status enhancement, convenience, usability, newness, customization, performance...)	Customers get faster repairs and quicker estimates. (HIGH)	High / Medium / Low
Customer Value: How much value do we deliver to the payer? (e.g., getting the job done, status enhancement, convenience, usability, newness, customization, performance...)	Repair shops reduce idle time and improve scheduling. (HIGH)	High / Medium / Low
Service Value (Revenue): To what degree will this application generate revenue?	Subscription or per-inspection fees for shops/insurance companies, and also improving operational efficiency and mechanic workflow (MEDIUM)	High / Medium / Low
Service Value (Non-revenue): How much non-financial value does this concept add to our service? (e.g. employee engagement, corporate social responsibility, raising awareness, etc)	Improving operational efficiency and mechanic workflow (MEDIUM)	High / Medium / Low
Adoption + Frequency of Use: How many new and current users will use this service, and how frequently?	Auto repairs are common and shops process many vehicles weekly (HI	High / Medium / Low
Risk to User/Stakeholder: How likely is this to cause harm to end-users or stakeholders? (e.g., their physical or psychological safety, civil liberties or rights, or economic opportunity)	Worst case is incorrect part suggestion, and the mechanic would verify anyway for flagged cases (LOW)	High / Medium / Low

Can This Be Built?		
Data Quality: Is there enough high-quality data to achieve 'good enough' model performance? (e.g., think about accuracy, biases, completeness ... of the data)	Large datasets already exist (HIGH)	High / Medium / Low
Minimum Performance: How well does the model need to perform for users to experience value?	Even being ~60-70% would save human time (MEDIUM)	Excellent / Good / Moderate
Data Bias: Is the data Biased? Will the system perform differently for certain groups or scenarios, potentially causing significant harm?	Damage detection is mostly object-based, not demographic-based (LOW)	High / Medium / Low
Error Frequency: How frequently would errors occur?	Images may be blurry, incorrect, or incomplete.	High / Medium / Low
Error Severity: How severe would an error be?	Human review prevents serious mistakes.	High / Medium / Low
Bad Actors: How easily could bad actors misuse the system to cause harm? (e.g., leading to harming data quality, security breaches, or monetary loss)	No one would submit the wrong photo of their car	High / Medium / Low

Should This Be Built?		
Build Costs: How expensive/time-consuming to build a dataset, develop a model, or fine-tune a pre-built model? (e.g., systems license, programming, data collection, employees, training, etc)	Use pre-trained vision models with some fine-tuning. (MEDIUM)	High / Medium / Low
Run Costs: How expensive would it be to run? (e.g., electricity, frequency of updating and retraining the model, data storage, computing power requirements, etc)	Unknown	High / Medium / Low
Societal Harm: How likely is it that this will indirectly cause societal harm? (e.g., job loss, sustainability, etc)	Primarily improves efficiency rather than replacing workers (LOW)	High / Medium / Low
Judgment		
This idea is a:	Keeper-Maybe	Keeper / Maybe / Show-Stopper

Risk Matrix



VOTE on final product choice

Terminology Simplifier: III
Construction Document Verifier: II
Car Damage Analysis: II
Edtech Character speak: I
Marketing image generation: I
Edtech tone indication:
Children's Storybook image generation:
Car Salesperson Matching:

Initial Capability Brainstorm - Alyssa

CAPABILITY: RENDER

- **Children's Story App**: Generate illustrations for each scene based on story descriptions. The app analyzes story descriptions and generates a series of illustrations.
- **Blog Thumbnail Generator**: Automatically generate blog or newsletter cover images from article titles. By scanning the article title or a brief summary, the AI generates a stylized cover image.
- **Presentation Slide Visual Generator**: Generate simple visuals or illustrations for slides from bullet points. **Restaurant Menu Visualizer**: Generate menu item images from dish descriptions for online ordering platforms.
- **Book Scene Visual Companion**: Readers generate visual interpretations of scenes from the book they're reading

CAPABILITY: VOCALIZE

- **Language Learning App**: Read pronunciation and example sentences aloud
- **Customer Support Voice Responses**: Convert scripted responses into natural voice messages
- **Job Interview Practice Tool**: Interview questions and feedback are read aloud to simulate real interviews.

Concept Cards - Alyssa

Concept Cards					
Description	Rendering pictures for children's storybooks				
Companies	Publishers of children's storybooks		What does the system do?	Given a storyline, renders images for different scenes for a children's storybook	
	Children's book authors				
	Educational Institutions		What data does the system have?	A storyline	
Users	Authors/Teachers			Sections of dialogue divided into scenes	
	Children			Descriptions of characters and background surroundings	
Payer	Authors/Teachers				

Concept Cards - Alyssa

Concept Cards					
Description	Rendering for Marketing				
Companies	Marketing sector	Generate ads instead of spending millions on filming and producing ads	**What does the system do?**	Given a description provided by marketing team, generates an image or video that displays the product value	
	Consumer Brands	Generates variations of product renders			
	Luxury Goods	Render custom orders for the customer to visualize before purchasing		A description of the product and what it does	
Users	Ad manager	Uses this new system as a tool to design ads efficiently while remaining cost effective	**What data does the system have?**	A creative vision and detailed description of what the final product should look like	
	Prompt Engineer	Generates prompts for the system to ensure the creative vision is met		Images of the product	
Payers	Brand Managers	Would pay for systems that improve brand image			
	Marketing Directors	Would pay for something that increases efficiency and decreases costs of marketing			

Initial Capability Brainstorm - Abdoul

CAPABILITY: Rank

- **Search Engine Results (SERPs):** Ordering web pages so the most "helpful" or "authoritative" links appear at the top for a specific query.
- **Product Recommendations:** Sorting an e-commerce catalog (Amazon, eBay) to show items a specific user is most likely to buy next.
- **Content Personalization:** Ranking news articles, TikTok videos, or Netflix movies based on a user's past engagement and "watch-time" probability.
- **Resume Screening:** Ranking job applicants for a recruiter based on how well their skills and experience match the job description.
- **Email Prioritization:** Identifying "High Priority" or "Focused" emails and moving them to the top of an inbox while burying newsletters.
- **Academic/Legal Retrieval:** Finding the most relevant case law or research papers in a database containing millions of documents.
- **Ad Tech:** Real-time ranking of advertisements to determine which one has the highest "click-through rate" (CTR) for a specific user session.

Initial Capability Brainstorm - Abdoul

CAPABILITY: Identify

- **Medical Diagnosis:** Identifying tumors in X-rays, fractures in MRIs, or specific skin conditions from dermatological photos.
- **Facial Recognition:** Identifying a specific individual to unlock a smartphone or tag a friend in a social media photo.
- **Fraud Detection:** Identifying suspicious patterns in credit card transactions that deviate from a user's "normal" spending profile.
- **Autonomous Driving:** Identifying pedestrians, traffic signs, lane markings, and other vehicles in real-time.
- **Sentiment Analysis:** Identifying the emotional tone of a customer review (Positive, Negative, or Neutral).
- **Spam & Malware Detection:** Identifying whether an incoming file or email contains malicious code or phishing links.
- **Manufacturing Quality Control:** Identifying defects, cracks, or missing components on an assembly line using computer vision.
- **Natural Language Entity Recognition (NER):** Identifying specific names, dates, locations, and organizations within a block of text.

Concept Cards - Abdoul

Concept Card					
Title					
Description	Translate into a description from computational code - Software/Corporate Documentation - Take in tooling or code created by workers and generate descriptions and formatting for the code in order to more seamlessly integrate it into larger codebases.				
Who are the...		*What should the system..*		*What do we know?*	
Payer (Customer)	Tech companies (startups and large enterprises) Engineering teams / CTO orgs	**Do / Act**	Parse codebases (functions, classes, dependencies) Generate: Inline comments README files API documentation Update documentation dynamically as code changes Suggest missing documentation or inconsistencies	**Data 1**	LLMs already perform strong code summarization (GitHub Copilot, GPT-based tools)
End User	Software engineers New hires / onboarding engineers Technical writers			**Data 2**	Large open-source repositories provide abundant training data for code + documentation pairs
Company or Department	Engineering teams DevOps / Platform teams Developer productivity teams	**Infer / Reason**	Documentation is consistently neglected due to time constraints Poor documentation slows onboarding and increases bugs Engineers prefer automated tooling over manual writing Companies lose productivity due to knowledge silos	**Data 3**	Static analysis tools can extract structure (ASTs) to improve accuracy of generated docs
Impacted Individuals	Developers (time savings) Technical writers (role shift toward editing) Product managers (clearer system understanding)			**Data 4**	Companies already pay for developer productivity tools (high willingness to pay)

Concept Cards - Abdoul

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End User	Software engineers New hires / onboarding engineers Technical writers			**Data 2**	Large open-source repositories provide abundant training data for code + documentation pairs
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Impacted Individuals	Developers (time savings) Technical writers (role shift toward editing) Product managers (clearer system understanding)			**Data 4**	Companies already pay for developer productivity tools (high willingness to pay)

Concept Assessment and Selection

For our original concept assessment, we looked at all the concept cards we had generated and ranked them on the risk benefit matrix.

We then voted on our favorite concepts, conferred with the professors, and moved forward with the one we found most interesting and both technologically and financially feasible (Smart Document Analysis).

We also based our decision off our own personal pain points (not many of us had experiences or desires with things like car rentals or building construction, but all of us had struggled with documentation before)

Financial Case

For the financials of this project, it was largely assisted by our working demo and AI. From the demo, we had found that the rough cost of one page was $0.03. We then also asked AI to help us scale this to the size of a larger business, analyzing longer documents, and rapid use. It gave us a similar number, estimating around $0.02 - $0.04 worth of tokens in cost per page. We also considered the time it took to create our rough demo to decide that, given 2 engineers for 3 months, they could complete this feature and fully integrate it into a larger service, or as a standalone product. We assumed the engineers were at $100,000 salary for this. We then decided that we should charge per-page instead of per-document, as some documents can be many tens or hundreds of pages long. At $0.10/page, this leaves us open to a lot of profit, of which we used to prove our ROI on the later slide.

Technical Case

The live demo showcased the feasibility of a baseline MVP of the front end. However, the real impact comes from the brains of our ideas, so to speak. We are not building a model from scratch, and yet we will be able to uniquely utilize public records to improve the accuracy of our service. The “Our Edge”-titled slide goes into more detail.

Consequence Scanning

We went through real-world examples, like medical treatment options, contract clauses, things people actually have to make decisions on, and we mapped out the possible outcomes. From there, we started thinking about what were the worst-case scenarios as well as the most likely scenarios. So we came to the conclusion that two risks that encapsulated the mix of those two scenarios were that AI could misinterpret a document and data privacy/security concerns. To figure out mitigations, we noted which parts of the document were high-stakes versus low-stakes, and which language was really tricky or ambiguous. That helped us figure out where the AI needed to focus its attention and how it should flag uncertainty for the user. For security, we knew we still needed industry-level protections, like AES-256 encryption, TLS for transfers, strong access controls, and multi-factor authentication, which are all standard for protecting sensitive information. That way, even if the platform was independent for now, user data stays secure.

Draft Pitch

Document Analysis Feature

Alyssa Zhu, Amy Zhang, Davis Germain, Abdoul Ndiongue, Jedidiah Yang

Problem

- Jordan is a 32 year old engineer
- He regularly encounters complex, high-stakes documents
    - Medical reports after doctor visits and legal contracts for housing, employment, or financial services
- Legal language is dense, ambiguous, and time-consuming to decode
- He needs context-aware explanations tailored to their specific document with high accuracy and reliability



Solution

Develop an AI-powered interpretation tool that converts complex medical and legal text into plain-language, context-aware explanations.



Key Features:

- Plain-language translation: Simplifies jargon into clear, concise explanations
- Contextual interpretation: Explains not just definitions, but what the information means for the user
- Real-time input: Users can ask questions, attach other documents or reports, or type notes for immediate clarification
- Privacy-first design: Ensures sensitive personal data is securely handled

NEXT

10. Each party hereby acknowledges and agrees that, in the event of any breach of this Agreement by the other party, including, without limitations, the actual or threatened disclosure of a disclosing party's Proprietary Information without the prior express written consent of the disclosing party, the disclosing party will suffer an irreparable injury such that no remedy at law will afford it adequate protection against or appropriate compensation for such injury. Accordingly, each party hereby agrees that the other party shall be entitled to specific performance of a receiving party's obligations under this Agreement as well as further injunctive relief as may be granted by a court of competent jurisdiction.

11. The term of this agreement is for two (2) years, commencing on the "Effective Date."

AGREED TO:

Social Security Number _____

SIGN

Signature _____

Printed Name David Guerrero

Date: 7/14/2014

Injunctive relief
Usually defined as ….., in the context of your **lease document** it refers to the case of ….

This means that if your house is destroyed in a tornado, your landlord can **xyz**

Document Summary and Analysis

This is a **lease agreement** for a **two year** period for a rental house on **xxyyzz Ave.**

What does section 2 of the lease cover? Are utilities included in my rent price? |

DocuSign Inc.

Technical Feasibility

- Leverages **existing** pre-trained language models for analysis
- Can **fine-tune** models on large datasets of legal documents if needed
- Natural language processing **easily** identifies convoluted or risky clauses
- Output **seamlessly integrates** into platforms like DocuSign



Financial Feasibility

Consider DocuSign:

Currently has ~1.7 million paying customers
>1 billion regular users

~2,000,000 signed up users

X $0.10 price per-page

~ $2,000,000 for **less than** 2% of current users with just **one 9-page document** scanned

User & Company Applicability

- All users will be able to comprehend their important documents (legal, medical, job, etc.) better, while this product can earn revenue with a per-use cost

Cost to Develop

- Large datasets of legal and other important documents already exist
- Builds upon pre-existing models and screen-overlay UI

Customer Retention

- This product will purely enhance user experience.
- Choice between a 0.10/page or $30/month price to reward power users

Desirability

For DocuSign:

- **Smoother User Experience**
- Trustworthy Rapport with Signers
- **Product Differentiation**

For Users:

- **Complex Terminology** that a user cannot understand without an extensive technical background
- Document **length** is intimidating and overwhelming
- **Privacy concerns** with asking for clarification about sensitive information from the internet, other people, or general LLMs
- Potential **deliberate intent to harm** hidden inside the contract, such as in the entertainment industry or predatory landlords

Risks

- **High-Stakes Misinterpretation**
- Incorrect translation of legal/medical language can lead to serious consequences (harmful contracts, legal risk).
- **Mitigation:** Flag uncertainty + show original vs. simplified text. Encourage expert consultation for high-stakes cases.



- **Data Privacy & Security Risks**
- Sensitive documents increase risk of breaches and privacy violations.
- **Mitigation:** Use secure platforms like DocuSign with encryption, access controls, and MFA.

FATE/FAIR

Fairness & Bias Control
Avoid using sensitive demographics (race, gender) unless required; account for bias in training data.

Transparency & Audit Trail
Maintain full context history with downloadable records for compliance.

Source Clarity & Disclosure
Label all outputs with sources, links, and AI-generated disclaimers.

User Control
AI assists only — never takes actions on the user's behalf.

Individual Crit Reflections (Davis)

For the draft pitches, I worked on the financial feasibility page. The biggest piece of feedback I got was that the slide was too cluttered. I was crowding the page with things like customer retention, which are apparent from the rest of our pitch. I was also reminded how important it is to have some back of the napkin math, however mine wasn't formatted properly or it needed to be more conservative to fit the rest of the pitch. I worked on reducing the amount of text on the screen to make the information more digestible, as well as adding the costs to a separate slide also showing build timeline. This way it was spread out more and the slides were more readable.

Individual Crit Reflections (Jed)

- My main focus was the Risks & Mitigations slide.
- Chose misinterpretation of high-stakes info and data privacy/security as the two top risks, since these could have serious real-world consequences for users.
- Developed practical mitigations: flagging uncertain sections, showing original + simplified text side-by-side, and outlining security measures we can implement without relying on DocuSign.
- Early versions of the slide were too wordy, so I worked on condensing the text while keeping the key points clear and digestible.
- Debated between pitching to DocuSign or pitching to VC, and went with VC, so had to learn how to do things in a VC-friendly way, having clear concise points and having things readable.
- Biggest takeaway: you need to show understanding of risks and solutions without overwhelming the audience, less is more.

Individual Crit Reflections (Abdoul)

I was not in attendance due to travel.

Individual Crit Reflections (Amy)

- We want a more interesting hook that still remains user-centered (this is a problem that affects everyone, not just a specific market of people)
- Desirability for the platform only needs to be considered if we are looking to get acquired, otherwise we don't need to include that section
    - Desirability for the users should be mostly self-explanatory as well, consider merging it in with a solutions capabilities slide. A well-written introduction can cover all the points across all three.
- Streamline ethical principles a little more, they're a bit repetitive
- Watch the timing, make sure what you're saying is supplemented by the slides, not a 1:1 with the text your audience can already read on their own
- Consider privacy/ethics concerns with sensitive or proprietary information
- I created the mid-fi mockup, the desirability, and FATE/FAIR analysis slides. I also modified and presented the tech feasibility during the practice pitch.

Individual Crit Reflections (Alyssa)

For the draft pitch, I worked on defining the problem and the solution. In the critiques we received a lot of feedback on how there was an excess of text on the slides and we were consistently crossing the 5 minute threshold. I was able to define the problem more clearly and make it more relatable by changing the customer persona to be a CMU student. The solution and desirability slides were similar, so we reformatted the solution slide to display the question we were trying to answer. I reduced the amount of text on both slides as well so that the slides would cover the basics and provide room for elaboration during presentation itself.

General Crit Reflections

- Need to work on timing
- Moving away from DocuSign platform brings the security risk, so we should think about containing it strictly within DocuSign
- Needs a better name
- Check if DocuSign already has features like this
- Start considering costs for running and creation
    - dynamic/per page pricing instead of by document
- Consider that Longer documents may be harder to read without being hallucinatory
- A lot of text/reading off slides
    - Reduce to bullets and talk it out in final pitch
- Add disclaimers
    - Someone may over rely on our product decisions
        - We should include a disclaimer that says this is AI generated
- Better hook, write a “how might we”

Solution from CRIT

The solution slide originally overlapped a lot with the desirability slide. During revisions we changed the slide to be a “how might we__?” statement to show the problem we were trying to solve and the value we wanted to bring to customers. We also were able to refine our solution and emphasize how the solution creates value feasibly. The critique also provided helpful feedback for how we could expand our target market by pitching this idea as an add-on for docusign. We were also able to ensure Docusign doesn’t have our proposed feature yet to ensure we’re co-creating value with this AI feature.

Repair of Inferences Errors

We clearly state on the MVP site that we are not meant to be a replacement for licensed legal professionals. Thus, in the odd case that there are inference errors in our tool, the outcome should be reviewed by a legal professional nonetheless, who should catch it. Perhaps a feedback system can be incorporated in the future so that any error such as this will be flagged for future cases.

ROI Calculation

For the ROI calculation, our demo was very helpful. It showed what we can expect from token costs, as well as how long this would take to build. From what we found at $0.03 per page, as well as with some assistance from AI in understanding how these costs would scale, we estimated that the maintenance costs were around $25,000 (given the customer population we assumed in the ROI calculation). We also added $75,000 to cover the initial build costs, as described in the final pitch (2 SWE Engineers @ $100k/year for 3 months). For our ROI, we made a semi-conservative estimate of 1,500,000 initial users (this number is based on the >1 billion users from 1 platform like DocuSign). These initial users would scan 1 9-page document, which is short for many legal documents, at $0.10 a page. This yields roughly $1,350,000, and subtracting the build costs led us to a rough **13x ROI.**

Avoidance of Ethical Issues

The FATE slide helped us formalize the ethical side of our tool. We clarified that the AI is never making decisions for the user, only providing interpretations and consequences of documents. During revisions, we focused on transparency and accountability. All AI-generated content is clearly labeled, sources are cited, and users can download a full record of the analysis. We explicitly addressed bias and fairness by stating that the system does not take into account things like gender, race, or socioeconomic status unless specifically instructed so that it mitigates unintended discrimination and ensures the tool is applied equally to all users. We reinforce professional user responsibility by making clear that the tool is there to help, not do for you. In addition to that, we show that thoughtful safeguards increase confidence in using the product for high-stakes decisions.

Final Pitch

V. STANDARD OF CARE

V.1. Receiving Entity shall use at least the same degree (but no less than a reasonable degree) of care and protection to prevent the unauthorized use, dissemination, tampering, or copying of any Confidential Information as Receiving Entity uses to protect its own Confidential Information. OMH has Policy and Standards to support the use and disclosure of each of the information classifications. If in doubt regarding the standard of care to be applied, the Receiving Entity can apply these to form the basis of their use and disclosure of OMH information assets.

V.2. Receiving Entity agrees to routine, unannounced inspections by the OMH, through its Information Security Office, to ensure the level of protection is maintained. To the extent practical and possible, such unannounced inspections shall occur during the normal business hours of Receiving Entity.

V.3. In the event of unauthorized access to Confidential Information, the Receiving Entity shall provide any and all assistance required by OMH to prevent further unauthorized access and to cooperate with OMH to secure the Confidential Information and protect OMH's proprietary rights.

VI. REPORTING OF UNAUTHORIZED DISCLOSURES AND OTHER INFORMATION SECURITY BREACHES

Any accidental or deliberate unauthorized disclosure of Confidential Information, information security breach, or suspicion of same, shall be immediately reported by Receiving Entity to the OMH Information Security Officer. The Receiving Entity shall maintain a written record of all such incidents.

VII. RECORD MAINTENANCE AND RETENTION

VII.1. Receiving Entity agrees to keep sufficiently descriptive and accurate records of data disclosures made under this Agreement, which shall identify the parties and the date of the DEA and/or CASA, as applicable, under which the data is disclosed.

VII.2. For each distinct purpose for which Receiving Entity wishes to obtain Confidential Information and/or access OMH Systems, a separate and distinct DEA and/or CASA, as applicable, shall be executed, which shall be made a part of and incorporated within this Agreement. If Receiving Entity wishes to obtain additional data or access to additional OMH Systems for the identical purpose as previously identified as the Purpose and Nature of Work in the original DEA and/or CASA, the original DEA and/or CASA may be amended upon mutual agreement of both parties to this Agreement. Once fully executed, the original copy of this Agreement and a copy of each DEA and/or CASA, or amended DEA and/or CASA, as applicable, shall be furnished to the OMH Information Security Officer for retention.

VIII. OWNERSHIP, INTELLECTUAL PROPERTY AND COPYRIGHT

VIII.1. It is a presumption that OMH retains full copyright and intellectual property rights over all its Information Assets and OMH Systems, wherever they may be stored.

VIII.2. OMH has legal ownership of the contents of all files stored on its OMH Systems as well as all messages transmitted via these systems. OMH reserves the right to access this information without prior notice whenever there is a genuine business need. Any use of OMH Systems requires OMH management approval and is subject

RESIDENTIAL LEASE AGREEMENT

[Single-Family House]

Click Here for Instructions

This Residential Rental Agreement ("Agreement") is entered into by and between _____ ("Tenant"), and _____ ("Landlord"). Landlord and Tenant are collectively referred to in this Agreement as the "Parties". This Agreement shall be effective as of the date executed by Landlord, as set forth below.

For the covenants contained herein, and other good and valuable consideration, the receipt and sufficiency of which is hereby acknowledged, the Parties agree as follows:

1. **PREMISES**: The leased premises shall be comprised of that certain personal residence (including both the house and the land) located at _____ ("Premises"). Landlord leases the Premises to Tenant and Tenant leases the Premises from Landlord on the terms and conditions set forth herein.

2. **TERM:** The term of this Agreement shall be a period of one (1) year, beginning on _____, and ending on _____. Any holding over after the expiration or earlier termination of the term without Landlord's prior written consent shall be a default of this Agreement and shall not be construed to be a tenancy from month to month, unless Tenant pays and Landlord accepts payment of rent for the next full calendar month (plus, if the term ends on a day other than the last day of the month, rent for the remainder of the month during which the term ends). If such payment of rent is so paid and accepted, this Agreement will automatically renew on a month to month basis, and will continue as such until terminated by either party in accordance with the following: written notice of termination is to be given by either party at least thirty (30) days before the designated date of termination, and the designated date of termination shall be the last day of a calendar month. If notice of termination is given, this Agreement shall terminate on the date for which notice is properly given. Except as otherwise set forth in this Agreement, all of the terms and conditions of this Agreement shall apply during any month to month tenancy.

3. **MONTHLY RENT:** The rent to be paid by Tenant to Landlord throughout the term of this Agreement is $_____ per month and shall be due on the 1st day of each month. Tenant shall pay a $**50.00** late fee for any rent not received by Landlord by the fifth (5th) day of the month. Tenant shall pay any returned check fees. All delinquent rents or other expenses due from Tenant shall accrue interest at the rate of 15% per year until paid. All interest shall be deemed additional rents. Rent for the first month (or, if applicable, partial month) of the term shall be paid to Landlord at the time this Agreement is executed. Rent for any partial month shall be prorated. Tenant shall not deduct or offset against rent unless expressly permitted by applicable law.

4. **UTILITIES:** To the extent permitted by applicable utility service providers, Tenant shall transfer all utility accounts into Tenant's name promptly upon taking possession of the Premises. Tenant shall pay, prior to delinquency, for all utilities (including, without



ContractClarity

pre-trained smart document analysis

Davis Germain, Abdoul Ndiongue, Jedidiah Yang, Amy Zhang, Alyssa Zhu





How might we assist users in understanding complex documents while maintaining trust and agency?





Capability



Simplify

- Context-aware definitions for complex terminology
- Securely ask private questions



Summarize

- Cut through to the key points of a clause or whole document



Identify Traps

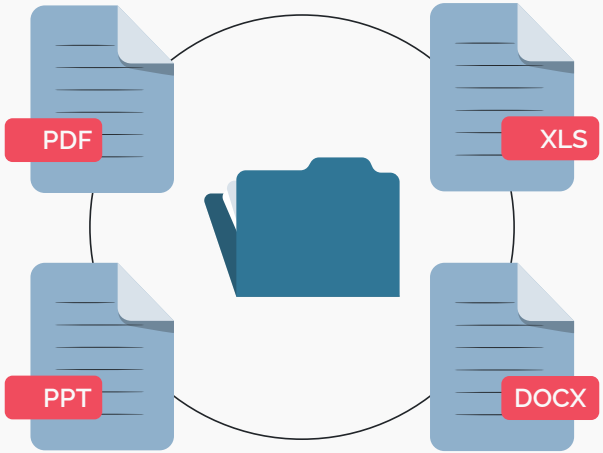
- Predatory terms
- Potential future pain points
- Saves consulting fees



Desirability

For the Platform

- Smoother user experience
- Builds a **trustworthy rapport** with signers
- **Product differentiation** in the market
- Provides useful data on how users interact with documents on the platform



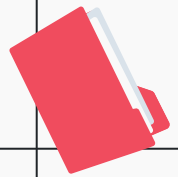
For Users

- Clarifies **complex terminology** without requiring a technical background
- Summarizes incredibly long documents
- Reduce **consulting or medical** fees while protecting privacy
- Catches intent to harm in predatory contracts

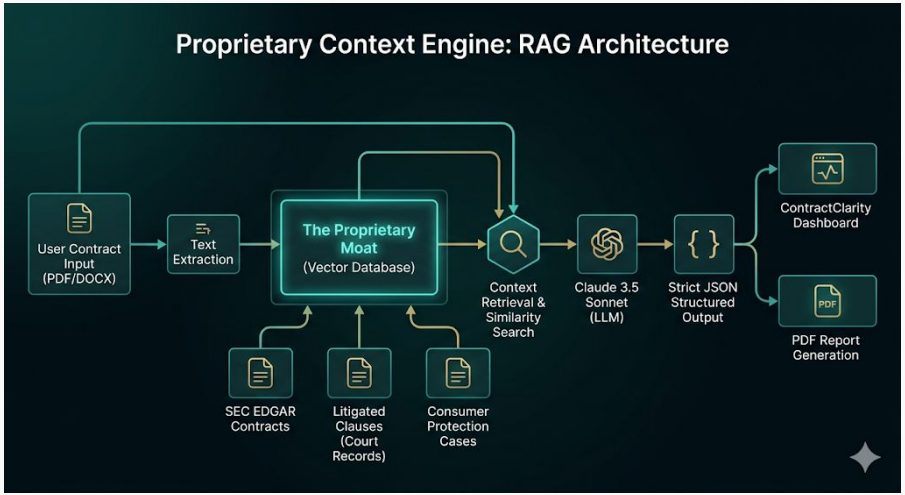
Live Demo

Why us?

Horizontal → Vertical



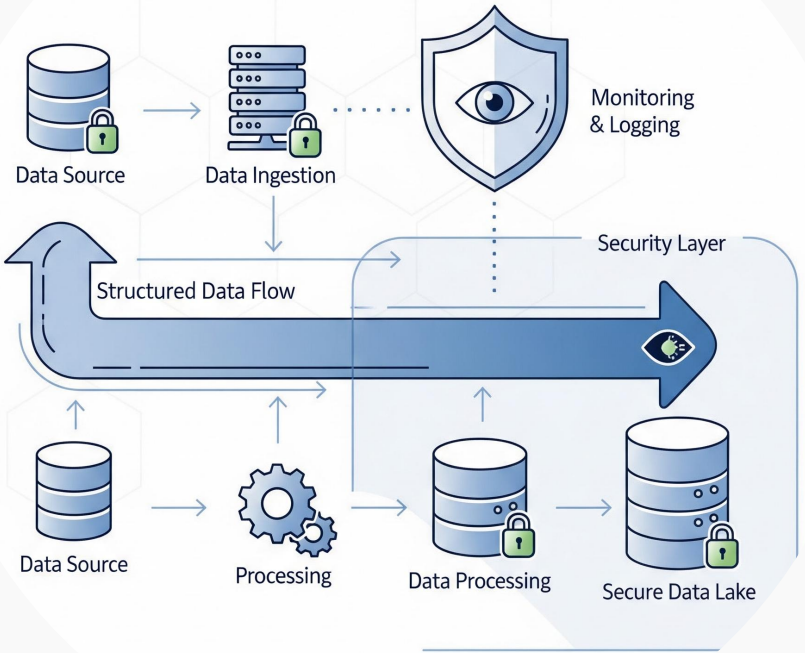
Our Edge: Deliberate Data



- **More than Generic AI:** We don't rely on just base-level LLM knowledge. We use Retrieval-Augmented Generation (RAG) to inject real, historical legal context into every analysis.
- **The "Litigation Database":** Our engine cross-references uploaded contracts against a growing vector database of historically litigated agreements, SEC EDGAR filings, and consumer protection cases.
- **Predictive Risk Scoring:** We don't just translate; we flag the exact phrasing that has historically led to exploitative outcomes, hidden fee disputes, or costly litigation.

Technical Feasibility

- Built on Next.js and React for modern, edge-ready infrastructure.
- Utilizes state-of-the-art LLMs constrained by strict structured outputs.
- Deterministic AI processing is forced to return highly specific JSON data.
- Documents are processed ephemerally for zero-retention privacy.
- Backend logic is decoupled for an extensible B2B API.



Build Timeline & Costs

01 $75K

Build Costs

2 Engineers
@ 100k Salary for 3 months

Coding Agents make it possible to use minimal human resources

02

Acquire Dataset

Including large datasets (i.e. court) that would be too much for Claude continuously

$100K

03 $25K

Service Maintenance

Build costs without labor

Token costs ($0.03/page)

Potential overhead

Financial Feasibility

Total Addressable Market

$7.68 Billion

Document Management Systems

Serviceable Addressable Market

>1 billion regular users
~3.22 billion in revenue

Docusign Platform

Serviceable Obtainable Market

~0.15% of users paying for
A 9-page document

1,500,000 users
X $0.10 per-page

= **$1,350,000**

For 0.015% of current DocuSign users paying for a 9-page document

13x ROI

Risks & Mitigation

High Stakes Misinterpretation

Incorrect translation of legal/medical language can lead to serious consequences (harmful contracts, legal risk).

Mitigation

Flag any uncertainty

Show original vs. simplified text

Encourage expert consultation for high-stakes cases.

Privacy & Security

Sensitive documents increase risk of breaches and privacy violations.

Mitigation

Use secure platforms with encryption, access controls, and MFA.



FATE Analysis



Fairness

- Do not consider user demographics when advising



Accountability

- Save all suggestions and definitions for records & compliance



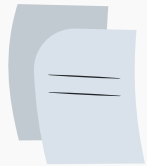
Transparency

- “AI-generated Content” Disclaimer
- Mark external sources



Ethics

- Never take action FOR the user
- Only educate or advise



Video Demo

Note: During the pitch there was a live demo. We unfortunately could not devise a video demo.

Individual Final Reflection (Davis)

Overall, I think this project fits very well into the previous work we have done in this class. With every project, we get one step further into the design process (i.e. initially it was a pitch, then P2 was a mid-fi mockup, now P3 is a demo). On top of this, we expand these skills into different areas (AUI, Pre-trained models, etc.). I think that beyond the scope of what we completed in this project, I will take the skills I have honed into future projects and the workforce hopefully, where I can be more confident in my pitching skills, ideation and innovation, and general understanding of making a pitch and what is necessary.

Individual Final Reflection (Jed)

I thought we did great. Apart from winning, I was proud of how we decided on the product, how we formulated a plan to make that product sellable, doing those steps to analyze and finalize that product, and successfully putting that info on the slides and pitching it. I think all of the steps we took along with good coordination allowed us to have a good result. This project taught me to think like real product designer. I mean, we were AI problem solvers that goes through the steps of brainstorming out a great product, thinking about how we will make this work to the best it can, and the amount of facets we have to consider to do that, and develop a product pitch like business people on Shark Tank. I love the mix of entrepreneurship and pure product design. I also liked that we could bring this project to life in a live-demo and see it hold up in practice. This project showed that things as simple as this could end up making a real-world impact, and I hope to have the chance to do that in the future.

Individual Final Reflection (Abdoul)

Honestly, looking back on this project, the most rewarding part was definitely moving past the spreadsheets and technical plans to actually build out the MVP. While the initial matchmaking and rapid assessment phases were great for filtering out the noise, seeing the model actually come to life in a demonstrable form made the whole pitch feel so much more real. It was a huge relief to see that the moderate model performance we planned for actually held up in practice, especially after obsessing over token costs and prompting strategies. In the end, showing off a working product that actually solved a stakeholder problem made all the pivoting and documentation feel totally worth it.

Individual Final Reflection (Amy)

I definitely found this project to be a more challenging working process than the previous one. Coordinating with my group to figure out what everyone was doing was also harder because of a few scheduling conflicts, so we didn't get as much time to practice as I would have liked. It was much more interesting to consider the potential legal ramifications of this project, since it was working much more intimately with a lot of sensitive data as opposed to my previous one.

The ability to create and live-demo a version of the tool was really cool, and developing and running the live-demo also gave me a lot of insight into what pre-trained models really can or cannot do currently, and helped me streamline the project more towards goals I knew were feasible.

I felt happier with my actual pitch this time, I think I worked better on my public speaking and audience hook.

Individual Final Reflection (Alyssa)

This project taught me a lot about the risks that come with pretrained models and how to mitigate those risks. The concept we came up with has a lot of high risk for hallucinations and for potential bias and ethical concerns. This project was less focused on the financials and more on how we can mitigate risk while demonstrating the value this project creates. I enjoy how every critique and project we learn a new aspect of design within AI that needs to be considered. In the case of this project, I think AI safety has always been everyone's biggest concern when it comes to designing and implementing new AI products, the considerations we made for risks and ethics plays a huge part in the real world. With the skills I've gained refining our project, I hope to apply them to any AI products I encounter in the future.

Thank you!

P3 Process Book:
Pre-Trained Models